Vectors and Vector-Valued Functions

ANSWER KEY

Section 1: 3D Vectors, Dot and Cross Products

1 Given $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 4, 2, -1 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$.

a. 3 ✓

b. 7

c. 13

d. -1

$\mathbf{u} \cdot \mathbf{v} = (2)(4) + (-1)(2) + (3)(-1) = 8 - 2 - 3 = 3.$

2 Find the cross product $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 1, 0, 2 \rangle$ and $\mathbf{v} = \langle 0, 3, -1 \rangle$.

a. $\langle -6, 1, 3 \rangle$ ✓

b. $\langle 6, -1, -3 \rangle$

c. $\langle -6, -1, 3 \rangle$

d. $\langle 6, 1, 3 \rangle$

$\mathbf{u} \times \mathbf{v} = \langle 0(-1) - 2(3), 2(0) - 1(-1), 1(3) - 0(0) \rangle = \langle -6, 1, 3 \rangle.$

3 Find the area of the parallelogram determined by $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 3, 4 \rangle$.

a. $\sqrt{89}$ ✓

b. 9

c. $\sqrt{73}$

d. 10

$\mathbf{u} \times \mathbf{v} = \langle 8, -4, 3 \rangle$. Magnitude $\|\mathbf{u} \times \mathbf{v}\| = \sqrt{64 + 16 + 9} = \sqrt{89}$.

Section 2: Derivatives and Integrals of Vector-Valued Functions

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4 Given $\mathbf{r}(t) = \langle t^2, \sin(2t), e^{3t} \rangle$, find $\mathbf{r}'(t)$.

a. $\langle 2t, 2\cos(2t), 3e^{3t} \rangle$ ✓

b. $\langle 2t, \cos(2t), e^{3t} \rangle$

c. $\langle t, 2\cos(2t), 3e^{3t} \rangle$

d. $\langle 2t, -2\cos(2t), 3e^{3t} \rangle$

Differentiate component-wise: $\mathbf{r}'(t) = \langle \frac{d}{dt}(t^2), \frac{d}{dt}(\sin 2t), \frac{d}{dt}(e^{3t}) \rangle = \langle 2t, 2\cos(2t), 3e^{3t} \rangle$.

5 Find $\int_0^1 \langle 2t, 3t^2, 4t^3 \rangle \, dt$.

a. $\langle 1, 1, 1 \rangle$ ✓

b. $\langle 2, 3, 4 \rangle$

c. $\langle 1/2, 1/3, 1/4 \rangle$

d. $\langle 1, 2, 3 \rangle$

Integrate component-wise: $\langle [t^2]_0^1, [t^3]_0^1, [t^4]_0^1 \rangle = \langle 1, 1, 1 \rangle$.

6 Find the unit tangent vector $\mathbf{T}(t)$ to $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$.

a. $\langle -\frac{3}{5}\sin t, \frac{3}{5}\cos t, \frac{4}{5} \rangle$ ✓

b. $\langle -3\sin t, 3\cos t, 4 \rangle$

c. $\langle -\frac{3}{4}\sin t, \frac{3}{4}\cos t, 1 \rangle$

d. $\frac{1}{5}\langle 3\cos t, 3\sin t, 4t \rangle$

$\mathbf{r}'(t) = \langle -3\sin t, 3\cos t, 4 \rangle$. $\|\mathbf{r}'(t)\| = \sqrt{9\sin^2 t + 9\cos^2 t + 16} = \sqrt{25} = 5$. Thus $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{5}$.

Section 3: Arc Length and Curvature of Space Curves

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7 Find the arc length of the circular helix $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t\rangle$ for $0 \le t \le 2\pi$.

a. 10π ✓

b. 5π

c. 20π

d. 6π

Speed $\|\mathbf{r}'(t)\| = \sqrt{(-3\sin t)^2 + (3\cos t)^2 + 4^2} = \sqrt{9+16} = 5$. Arc length $L = \int_0^{2\pi} 5\,dt = 10\pi$.

8 State the formula for curvature $\kappa(t)$ of a space curve in terms of $\mathbf{r}'$ and $\mathbf{r}''$.

a. $\kappa(t) = \dfrac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$ ✓

b. $\kappa(t) = \dfrac{\|\mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^2}$

c. $\kappa(t) = \dfrac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{\|\mathbf{r}'(t)\|^3}$

d. $\kappa(t) = \|\mathbf{r}''(t)\|$

The standard vector formula for the curvature of a twice-differentiable space curve is $\kappa = \dfrac{\|\mathbf{r}' \times \mathbf{r}''\|}{\|\mathbf{r}'\|^3}$.

9 Find the curvature of a circle of radius R.

a. $\kappa = 1/R$ ✓

b. $\kappa = R$

c. $\kappa = 2\pi R$

d. $\kappa = 1/R^2$

A circle of radius R has constant curvature equal to the reciprocal of its radius: $\kappa = 1/R$.

Section 4: Velocity, Acceleration, and Projectile Motion

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10 If position is $\mathbf{r}(t) = \langle t^2, 4t, -16t^2 + 64t \rangle$, find the velocity and acceleration vectors at $t = 1$.

a. $\mathbf{v}(1) = \langle 2, 4, 32 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$ ✓

b. $\mathbf{v}(1) = \langle 1, 4, 48 \rangle, \mathbf{a}(1) = \langle 2, 0, -16 \rangle$

c. $\mathbf{v}(1) = \langle 2, 4, 0 \rangle, \mathbf{a}(1) = \langle 2, 0, -32 \rangle$

d. $\mathbf{v}(1) = \langle 2, 4, -32 \rangle$

$\mathbf{v}(t) = \mathbf{r}'(t) = \langle 2t, 4, -32t + 64 \rangle$. At $t = 1$: $\mathbf{v}(1) = \langle 2, 4, 32 \rangle$. $\mathbf{a}(t) = \langle 2, 0, -32 \rangle$.

11 Find the speed at $t = 0$ for a particle with velocity $\mathbf{v}(t) = \langle 3\cos t, 4, 3\sin t \rangle$.

a. 5 ✓

b. 7

c. 4

d. $\sqrt{7}$

Speed is the magnitude of velocity: $\|\mathbf{v}(0)\| = \sqrt{3^2 + 4^2 + 0^2} = \sqrt{9 + 16} = 5$.

12 Decompose acceleration into tangential and normal components: $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$. What is a_T?

a. $a_T = \frac{\mathbf{v} \cdot \mathbf{a}}{\|\mathbf{v}\|}$ ✓

b. $a_T = \frac{\|\mathbf{v} \times \mathbf{a}\|}{\|\mathbf{v}\|}$

c. $a_T = \|\mathbf{a}\|$

d. $a_T = \frac{\mathbf{a}}{\|\mathbf{v}\|}$

The tangential component of acceleration represents the rate of change of speed: $a_T = \frac{d}{dt}\|\mathbf{v}\| = \frac{\mathbf{v} \cdot \mathbf{a}}{\|\mathbf{v}\|}$.

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